Harbor Black (Mega Port / Naval + Ground + Air)

Executive Summary

Harbor Black is a **mega-port combined-arms battlefield** built for **naval warfare, ground combat, air superiority, logistics disruption, and autonomous multi-domain coordination**. The terrain features massive cargo terminals, multi-level container stacks, ship berths, drydocks, cranes, refineries, elevated highways, rail spurs, and fortified control towers. The map emphasizes Obsidian Protocol's autonomous port-infrastructure routing, sensor-fusion under industrial clutter, comms-relay warfare, and logistics-driven strategic play across sea, land, and air. In AR (Nreal Light/Air), the map anchors at ~4–5 m viewing distance, covering a 10x10 m physical area representing ~1 km² of mega-port terrain. Key features include naval berths, elevated overwatch, container-maze ground combat, air corridors, and multi-domain chokepoints.

Design Goals & Gameplay

Naval + Ground + Air Integration

Harbor Black prioritizes simultaneous multi-domain combat: ship-to-shore fire, drone reconnaissance, armored pushes through container yards, and air strikes from above.

Logistics Warfare

Cargo terminals, refineries, and rail yards serve as logistics hubs. Destroying or capturing them affects reinforcement flow and resource availability.

Container-Maze Ground Combat

Stacked containers create dense cover, ambush lanes, interior corridors, and unpredictable sightlines.

Air Superiority & Vertical Control

Aircraft, VTOLs, and drones dominate the vertical dimension. Air corridors between cranes and control towers define recon, interception, and strike patterns.

Autonomy & Infrastructure-Aware Routing

Autonomous units choose routes based on container density, crane hazards, water depth, elevated road stability, and sensor clarity.

Sensor & Information Warfare

Smoke, steam, metallic clutter, water reflection, and industrial noise degrade sensors. Relay towers on control towers restore clarity. Destroying relays creates blind pockets across the port.

Deployment & Progress

A fixed deployment budget (~10,000 points) forces players to choose between naval dominance, air superiority, or ground-based logistics disruption. Blue deploys from the southwest cargo terminal; Red deploys from the northeast refinery command.

Spatial Constraints (Nreal AR)

Device Capabilities

Nreal Light/Air: ~53° diagonal FOV, 1920×1080 per-eye resolution. Optimal viewing distance ~4–5 m.

Playable Area & Scaling

10×10 m physical area ≈ 1 km² mega-port zone (1:100 scale). Designed for tabletop AR.

Tracking & Occlusion

Inside-out tracking detects horizontal planes. Depth-mesh (if available) provides limited occlusion. Virtual objects render over real-world view; assume minimal real-world occlusion.

Layout Overview

Coordinate System

1000×1000 grid (1 unit ≈ 1 m). Origin (0,0) = southwest corner.

Key Locations

Blue HQ (Cargo Terminal Command) — (0,0), 150×150 m Cranes, containers, amphibious staging, drone pads.
Red HQ (Refinery Command Spire) — (1000,1000), 150×150 m Refinery tower with radar mast, AA emplacements, and sensor arrays.
Central Berth & Drydock Zone — (500,450), 300×300 m Ship berths, drydocks, naval combat, amphibious landings.
Container Yard Cluster — (250,800), 120×120 m Stacked containers, interior corridors, ambush lanes.
Elevated Highway Network — (800,300), 200×200 m High-ground overwatch, sniper nests, air-corridor control.
Harbor Crossing Corridor — (0,450 → 400,900) Primary chokepoint: bridges, water hazards, industrial clutter.
Service Road Route — (300,0 → 450,350) Safer but predictable; vulnerable to overwatch from cranes and towers.
Control Tower Relay A — (650,150), +240 m Long-range sensor vantage point.
Control Tower Relay B — (150,650), +260 m Comms relay hub and logistics-monitoring node.
Neutral Forward Outpost — (500,950) Drone resupply zone.

Chokepoints & Sightlines

Harbor Crossing Corridor

Major chokepoint; controlling bridges determines movement between port sectors.

Container Yard Cluster

Dense interior flanking routes; unpredictable cover and ambush potential.

Berth & Drydock Zone

Naval + ground + air engagements; water reflection affects accuracy.

Elevated Highway Network

Long-range overwatch; ideal for air-ground coordination.

Sightlines

Control Tower Relay A provides long-range sensor coverage (~350 m). Smoke and water reflection create fog-of-war pockets.

Deployment Zones

Blue Deployment: SW cargo terminal + amphibious staging.
Red Deployment: NE refinery command + AA emplacements.
Neutral Forward Zone: Outpost at (500,950) for drone drops.

Assets & Environment

Terrain

Berths, docks, drydocks, container yards, refineries, elevated highways.

Vegetation

Sparse industrial weeds, riverbank grass.

Buildings & Props

Military: AA guns, radar dishes, bunkers. Industrial: Warehouses, cranes, pipelines, refineries. Logistics: Trucks, crates, fuel drums, boats, drones.

Effects

Smoke, steam, water spray, sparks, industrial noise.

Audio

Machinery, water flow, distant alarms, radio chatter.

LOD & Budget

Asset Cat	Count	Tri/Item	Total Tris
Port/Indu	3	200k	600k
Industrial	4	80k	320k
Buildings/	15	50k	750k
Roads/Pa	4	30k	120k
Props	50	10k	500k
Total	—	—	**~2.29M**

Texture Budget

2048² for terrain/buildings, 1024² for props. DXT1/5 compression. Target ≤100 MB.

Unreal / Nreal Integration

Engine & Plugins

Unreal ARTemplate + Nreal SDK (XREAL). ARKit/ARCore enabled.

Rendering

Forward renderer, stationary lights, baked GI, aggressive culling.

Anchors

Single spatial anchor for entire map.

Interaction

Phone pointer for pan/zoom; HUD as screen-space widgets.

Multiplayer

Client-server architecture; offload AI/physics to server.

Implementation Plan & Testing

Milestones

1. **Prototype Layout — Month 1** Block-out berths, container yards, elevated highways, AR anchor test.
2. **Core Systems — Month 2** NavMesh (land + water), multi-domain autonomy, industrial hazard logic.
3. **AR Integration — Month 3** Anchors, plane detection, basic interactivity.
4. **Content Fill — Month 4–5** Final models, textures, lighting, audio.
5. **Gameplay & Polish — Month 6–7** Objectives, balance, VFX (smoke, steam, water spray).
6. **Testing & QA — Month 8** Multiplayer stress test, sensor validation.

Testing Plan

Autonomy

Verify amphibious routing, container-maze navigation, air-ground coordination.

Comms/Sensors

Test relay destruction, fog-of-war updates, LOS checks under industrial smoke.

Performance

≥60 Hz AR render on target device.

QA Checklist

Navigation, multi-domain logic, deployment budget enforcement, anchor drift, sensor accuracy.